Airbnb Project Summary

Airbnb Multi-City Analytics — Project Summary

1. Overview

End-to-end analytics project on real, unprocessed Airbnb data across 4 cities — New York, London, Paris, Barcelona. Covers Python cleaning → MySQL ETL → Power BI dashboard.

Why this dataset: Inside Airbnb is real, publicly available scraped data — not a pre-cleaned dataset — allowing the project to demonstrate genuine data engineering work (cleaning, modeling, aggregation) rather than just chart-building.

Scale: ~215,869 listings | ~78M calendar rows | 4 cities

City	Listings
London	92,638
Paris	77,679
NYC	30,259
Barcelona	15,293

2. Architecture

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Inside Airbnb (raw CSV) → Python (csv module, cleaning) → MySQL 8.0
  (staging → cleaned/typed → aggregated → views) → Power BI (model, DAX, dashboard)
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All cleaning, type-casting, and business logic is handled in SQL before reaching Power BI — keeping the BI layer focused purely on modeling and visualization.

3. SQL Layer — What Was Built

Object	Purpose
`listings_staging`, `calendar_staging`	Raw CSV import, all columns as TEXT
`listings_clean`	Fully typed and cleaned — prices, dates, counts cast to proper types with defensive validation
`calendar_monthly_agg`	78M daily calendar rows aggregated to ~2.8M monthly rows per listing
`vw_calendar_occupancy` / `vw_calendar_availability`	Monthly occupancy % and availability % per listing
`vw_listings_annual_occupancy`	Listing-level annual average performance, joined with price/room type/neighbourhood
`vw_rank_listings_occupancy`	Listings ranked by occupancy within each city (window function)
`vw_listings_monthly_revenue`	Estimated monthly revenue per listing (price × booked days)

Techniques used: staging/clean/aggregate/view layering, defensive type-casting with pattern validation, conditional aggregation (`SUM(CASE WHEN...)`), multi-table joins, window functions (`DENSE_RANK()`

`OVER (PARTITION BY ...)`), indexing for performance.

4. Power BI Layer — What Was Built

- Connected Power BI Desktop to MySQL; imported `listings_clean` and `calendar_monthly_agg`
- Built a star-schema relationship: `listings_clean` (1) → `calendar_monthly_agg` (many) on `listing_id`
- **DAX measures:** `OccupancyRate`, `AvailabilityRate` (via `DIVIDE`), `EstimatedRevenue` (via `SUMX` + `RELATED`), `TotalListings` (via `DISTINCTCOUNT`), `AvgPrice`
- **Calculated columns:** `SuperhostLabel`, `HostExperienceGroup` (via `SWITCH(TRUE(), ...)` bucketing)
- **3-page report:**
  - **Overview** — KPI cards, market map, city comparison charts, revenue trend over time
  - **Market Deep Dive** — room type and property type segmentation, top/bottom neighbourhoods by price, price-vs-occupancy scatter plot
  - **Host & Performance Analysis** — top 10 listings by revenue, superhost vs. non-superhost comparison, host experience vs. performance comparison

5. Business Findings

Page 1 — Overview

City	Occupancy	Avg Price
Paris	61.25%	$321.23
London	56.20%	$271.47
NYC	49.01%	$278.23
Barcelona	41.05%	$240.62

Paris leads on both price and occupancy simultaneously — unusual, since higher price would normally suppress demand, suggesting either supply constraint or strong, price-inelastic tourism demand. Barcelona lags on both — a possible sign of an oversupplied or softening market. Revenue peaks mid-summer across all cities before tapering into Q4.

Page 2 — Market Deep Dive - 73% of listings are Entire home/apt; “Entire home” specifically commands the highest average price ($442) even within that category. - Price varies roughly 20x by neighbourhood ($65–$1,384/night) — city-level averages mask substantial internal variation. - No clean relationship between price and occupancy at the neighbourhood level — price alone doesn’t explain demand differences.

Page 3 — Host & Performance - Genuine top 10 listings by revenue are premium Paris/NYC properties in the $3,300–4,400/night range. - Superhosts show slightly *lower* occupancy than non-superhosts (52.24% vs. 56.83%) at nearly identical pricing. - Host experience shows a non-linear relationship with performance — occupancy peaks at 6–10 years (65.57%), then dips for 10+ year hosts, who price higher instead.

6. Unified Business Insight

> *Performance in this market is driven by neighbourhood- and listing-level factors, not by the surface-level signals a business would typically reach for first — city label, price alone, Superhost badge, or host tenure. Any strategy built purely on these signals would likely underperform; real pricing and investment decisions need to happen at a more granular level than city or host-status averages can support.*

7. Known Data Limitations

- **Occupancy ≠ confirmed bookings** — the calendar marks a date “unavailable” whether it was booked by a guest or manually blocked by the host; the two can’t be distinguished in this dataset.
- **Calendar data is forward-looking** (~12 months from the scrape date) — availability further from the scrape date is less reliable, since hosts haven’t fully updated it yet. The Overview page is filtered to the near-term window for this reason.
- **Revenue figures are estimates** (`nightly price × booked days`) — not actuals, and don’t account for discounts, fees, or cancellations.

8. “Tell Me About This Project” (60-second version)

*“I built an end-to-end analytics project on real Airbnb data across four global cities — about 216,000 listings and 78 million calendar records. I used Python to clean the raw CSVs, then built a layered SQL pipeline in MySQL — staging, a cleaned and properly-typed layer, a monthly aggregation layer, and analytical views using joins and window functions for occupancy, rankings, and revenue. On top of that, I built a 3-page Power BI report with DAX measures covering occupancy, revenue, and host performance.

The findings were genuinely interesting — Paris leads on both price and occupancy simultaneously, which is unusual since those normally trade off against each other, while Barcelona lags on both. At the neighbourhood level, price doesn’t cleanly predict occupancy, and Superhost status didn’t correlate with higher occupancy either — both findings pushed back on assumptions I expected the data to confirm. The overall takeaway is that simple, city- or badge-level signals aren’t reliable enough for real pricing or investment decisions in this market — the real drivers sit at a more granular level.”*